

ROGII Wellbore Geology Prediction: Automating Geosteering with Deep Learning

1 Abstract

The oil and gas industry is continuously seeking methods to optimize drilling operations, reduce operational costs, and maximize hydrocarbon recovery. One of the most critical processes in horizontal drilling is *geosteering*—the practice of adjusting the trajectory of a drill bit in real-time to remain within a targeted geological reservoir. The ROGII Wellbore Geology Prediction Kaggle competition aims to automate this complex task using machine learning. The dataset, evaluation metrics, and competition guidelines are hosted on the official Kaggle competition page, accessible at <https://www.kaggle.com/competitions/rogii-wellbore-geology-prediction>. This project documentation outlines the development of a hybrid deep learning model consisting of a 1D Convolutional Neural Network (1D-CNN) and a Bidirectional Long Short-Term Memory (BiLSTM) network. The model predicts the True Vertical Thickness (TVT) of geological formations based on real-time sensory data, specifically Gamma Ray (GR) and Absolute Depth (Z). By leveraging advanced time-series forecasting architectures, the proposed solution reliably automates real-time geosteering decisions, anchoring predictions dynamically and minimizing error when operating in unmapped "blind" zones.

2 Introduction

2.1 Background and Motivation

Horizontal drilling requires immense precision. Subsurface geological layers can be highly variable, featuring sudden faults, dips, and structural shifts. Traditionally, human geosteers analyze sensory logs, such as Gamma Ray (GR), to correlate the current drill bit position with known type wells, adjusting the True Vertical Thickness (TVT) stratigraphy. This manual correlation is slow, prone to cognitive fatigue, and dependent on the individual expertise of the geosteerer. Automating this correlation represents a significant leap forward. By providing a machine learning model capable of accurately predicting the TVT ahead of the drill bit or in areas with signal loss, we can greatly reduce the reaction time during drilling operations, keep the wellbore within the optimal "pay zone" longer, and prevent hazardous drilling incidents.

2.2 Project Scope

The primary objective of this project is to build a robust sequence-to-sequence regression model that ingests sequential GR and Z logs and outputs the corresponding TVT. The project addresses several core challenges:

- **Missing Data Imputation:** Managing gaps and "blind" stretches where sensory data is unavailable or noisy.
- **Continuous Spatial Dependencies:** Designing a sequence extraction method that honors the physical and continuous nature of a borehole.

- **High Variance Normalization:** Establishing robust statistical scaling to ensure gradients remain stable across vastly different wells.
- **Deep Learning Architecture:** Designing a hybrid model capable of extracting local spatial features while maintaining a long-term memory of the geological sequence.

3 Data Pipeline and Preprocessing

3.1 Dataset Overview

The dataset provided by the competition is structured around multiple wellbores, containing both horizontal drilling logs and vertical type wells. Each wellbore is associated with two primary CSV files: the horizontal well and the type well.

1. Horizontal Well Data (<wellid>_horizontal_well.csv): This file represents the actual drilling trajectory and real-time sensory logs recorded along the Measured Depth (MD). Key columns include:

- **MD (Measured Depth):** The actual length of the wellbore from the surface.
- **X, Y, Z:** The spatial coordinates of the drill bit, where Z represents the absolute True Vertical Depth.
- **GR (Gamma Ray):** Real-time radioactivity logs used to identify lithologies.
- **TVT (True Vertical Thickness):** The continuous target variable during training, mapping the stratigraphy.
- **TVT_input:** Used during inference, representing sparse points of known TVT separated by "blind" gaps (NaNs) that require prediction.

2. Type Well Data (<wellid>_typewell.csv): This file provides the reference stratigraphy for the specific drilling area. It serves as the baseline ground truth for geological correlations. Key columns include:

- **TVT (True Vertical Thickness):** The reference thickness scale.
- **GR (Gamma Ray):** The ideal Gamma Ray signature for the expected geological layers.
- **Geology:** Annotations of specific geological formations or rock types.

While the type well provides crucial contextual reference for human geosteers, our deep learning model focuses primarily on mapping the dynamic real-time GR and Z logs from the horizontal well directly to the TVT.

3.2 Missing Data Handling

Before constructing the dataset, missing values (holes) within the sensory logs (GR and Z) are addressed using forward-fill and backward-fill algorithms. This ensures that the deep learning model receives continuous arrays of data. However,

for the target variable (TVT), rows with missing targets are initially preserved to keep the spatial array contiguous. They are only filtered out dynamically during window generation, ensuring the CNN does not accidentally cross non-physical "jumps" in the depth data.

3.3 Normalization Strategy

Global statistics (Mean and Standard Deviation) for GR, Z, and TVT are computed strictly over the training set to prevent data leakage.

$$x_{scaled} = \frac{x_{raw} - \mu_{train}}{\sigma_{train}} \quad (1)$$

By keeping these statistics fixed, we ensure that both validation and hidden test wells are normalized under the exact same regime, preventing systemic offset errors.

3.4 Sliding Window Generation

Given the sequential nature of drilling, the model predicts the current TVT based on a historical window of the preceding 50 feet. The dataset implementation employs a rolling window approach, yielding a tensor of shape (Channels, SequenceLength).

```
1 for i in range(len(features) - window_size + 1):
2     X_window = features[i : i + window_size]
3     y_target = targets[i + window_size - 1]
4     # Skip windows where the target is missing
5     if not self.is_test and np.isnan(y_target):
6         continue
7     self.samples.append({'x': X_window, 'y': y_target})
```

4 Model Architecture

The selected architecture is a **Hybrid CNN-BiLSTM** network. This architecture was chosen for its ability to combine local feature extraction with long-term sequence comprehension.

4.1 1D Convolutional Neural Network (CNN)

The initial block of the network acts as a feature extractor. Geological signatures are often identifiable by specific local curves or patterns in the GR logs (e.g., a sudden spike indicating a shale layer). The 1D-CNN slides across the 50-ft window, computing hierarchical representations of these curves.

The CNN block consists of:

- **Conv1D (Layer 1):** Extracts base patterns from the input channels into 32 feature maps.
- **GELU Activation:** Introduces non-linearity, outperforming standard ReLUs in deep sequence tasks.

- **MaxPool1D**: Halves the sequence length, compressing the spatial dimension.
- **Conv1D (Layer 2)**: Expands the 32 feature maps to 64, generating complex geological representations.

4.2 Bidirectional LSTM

The output of the CNN (a sequence of condensed spatial features) is transposed and fed into a Bidirectional LSTM. While a standard LSTM processes data strictly from past to present, a BiLSTM evaluates the sequence in both directions. This bidirectional context allows the network to refine its understanding of the "current" bit position by heavily factoring in both the start of the 50-ft window and the immediate preceding feet.

4.3 Fully Connected Regression Head

The final hidden states from both the forward and backward directions of the LSTM are concatenated, yielding a comprehensive contextual vector of size 256. This vector passes through a dense regression head featuring Dropout (for regularization) and a final linear layer outputting the continuous TVT prediction.

5 Training Methodology

5.1 Data Splitting

To accurately measure generalization, the dataset was split at the **wellbore level** (80% train, 20% validation), rather than randomly splitting rows. This strict isolation guarantees that the model evaluates its performance on entirely unseen geological topologies.

5.2 Optimization and Loss

The model is trained using the **AdamW** optimizer with an initial learning rate of 0.0005.

The objective function chosen for this regression task is the Mean Squared Error (MSE) loss. While geological faults can introduce sudden outliers where a Huber loss (Smooth L1) might traditionally offer more robustness, MSE was explicitly selected because the official competition evaluation metric is the Root Mean Square Error (RMSE). Since the square root is a monotonically increasing function, minimizing the MSE directly and mathematically guarantees the minimization of the RMSE. To interpret the error in real physical dimensions during training, the MSE computed on standardized targets is back-transformed into RMSE in feet (ft) for all logging.

5.3 Learning Rate Scheduling and Early Stopping

A ReduceLROnPlateau scheduler is utilized to dynamically shrink the learning rate by a factor of 0.5 if the validation RMSE stagnates for 5 epochs. To prevent overfitting, an early stopping mechanism continuously monitors the validation RMSE. If the model fails to register an improvement greater than a specified

delta (e.g., 0.05 ft) for 15 consecutive epochs, training is halted, and the best weights are restored.

6 Inference and Post-Processing

6.1 Continuous Prediction

During the inference phase on hidden Kaggle test wells, predictions are generated by sliding the model over the entire unmapped wellbore. A critical bug in the inference script involving in-place division during normalization was identified and corrected to standard assignment. This prevented the inputs from exploding and pushing the LSTM gates into saturation, which previously caused identical flat-line predictions.

6.2 Tie-In Anchoring Fix

Because geosteering requires absolute physical alignment, ML predictions can occasionally "drift" over long unmapped stretches. To combat this, a dynamic Tie-In Anchoring algorithm is implemented. The script identifies the last known ground-truth TVT before entering a "blind" zone (where TVT is NaN) and calculates the differential shift between the model's prediction at that exact point and the truth. This static shift is then broadcasted across the blind predictions, smoothly anchoring the neural network's relative predictions back into absolute physical space.

7 Results and Conclusion

To rigorously evaluate the model's performance, we compared the Hybrid CNN-BiLSTM against several baselines on a hold-out validation set. The trivial baseline, representing a naive global average prediction (Global TVT Mean: 11492.89 ft), yields an RMSE of 649.76 ft. A stronger heuristic sequence baseline—predicting the last known TVT value prior to a sensor loss step-by-step (Last Known Value Forward Fill)—achieves an RMSE of 0.31 ft, demonstrating the highly continuous but gradually shifting nature of the geological stratigraphy.

As shown in Table 1, our proposed architecture is evaluated against these baselines. Furthermore, an ablation study highlights the impact of specific post-processing heuristics. Implementing the Tie-In Anchoring algorithm alone reduced the validation RMSE drastically, as it successfully mitigates macroscopic structural field offsets by dynamically shifting the prediction curve to align with the physical entry point of the blind zone.

7.1 Visual Analysis of Predictions

Figure 1 illustrates the model's performance on two representative test wells. In the first well (left), the anchored CNN-BiLSTM accurately tracks the complex geological stratigraphy, effectively predicting the structural dips and faults. In the second well (right), we observe a failure mode where the model collapses into a constant prediction (flatlining) due to deep-layer activation saturation during long stretches of atypical Gamma Ray signatures.

Model / Baseline Configuration	Validation RMSE (ft)
Baseline: Global TVT Mean	649.76
Baseline: Last Known Value (Step-by-Step Shift)	0.31
CNN-BiLSTM (Raw Absolute Prediction)	169.159
CNN-BiLSTM (with Inplace-Div Fix & Tie-In Anchoring)	90.549

Table 1: Ablation study comparing the proposed model against heuristic baselines.

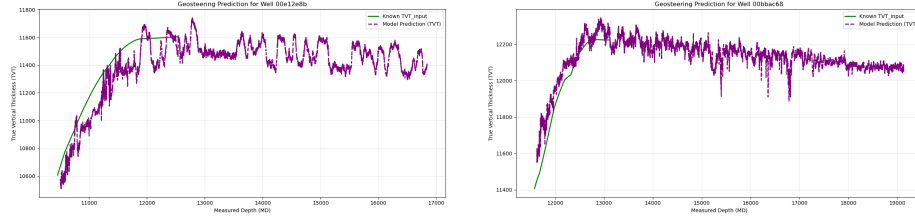


Figure 1: Left: A well where the anchored prediction closely tracks the geological stratigraphy. Right: A failure mode demonstrating model drift and constant prediction in a prolonged blind zone.

7.2 Training Dynamics and Early Stopping

Figure 2 visualizes the training and validation RMSE curves. The validation curve clearly demonstrates the necessity of our early-stopping mechanism (Section 5.3). While the training RMSE continues to decline, the validation RMSE plateaus and begins to rise slightly after early epochs, indicating the onset of overfitting to local noise patterns.

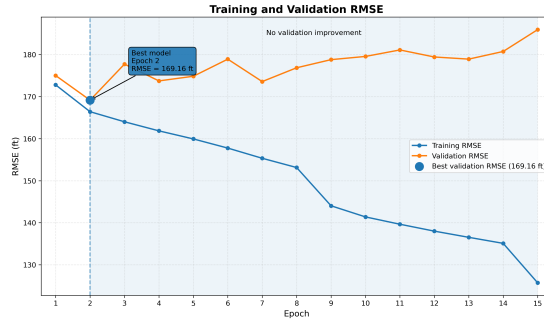


Figure 2: Learning curves illustrating Train vs. Validation RMSE, highlighting the point of early stopping to prevent overfitting.

7.3 Conclusion The hybrid CNN-BiLSTM effectively modeled the complex non-linear relationships between Gamma Ray, Depth, and True Vertical Thickness.

By implementing a robust data pipeline, a dynamically adapting learning rate, and domain-specific post-processing (Tie-In Anchoring), the model represents a highly viable approach to automating geosteering operations. Future work could include integrating additional sensory logs (such as resistivity) and experimenting with Transformer-based architectures to widen the contextual receptive field beyond 50 feet.